

Chapter 48

Neuroepigenetics and addiction

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Abstract

Drug addiction involves long-term behavioral abnormalities that arise in response to repeated exposure to drugs of abuse in vulnerable individuals. It is a multifactorial syndrome involving a complex interplay between genes and the environment. Evidence suggests that the underlying mechanisms regulating these persistent behavioral abnormalities involve changes in gene expression throughout the brain's reward circuitry, in particular, in the mesolimbic dopamine system. In the past decade, investigations have begun to reveal potential genes involved in the risk for addiction through genomewide association studies. Additionally, a crucial role for epigenetic mechanisms, which mediate the enduring effects of drugs of abuse on the brain in animal models of addiction, has been established. This chapter focuses on recent evidence that genetic and epigenetic regulatory events underlie the changes throughout the reward circuitry in humans, as well as animal models of addiction. While further investigations are necessary, a picture of genetic and epigenetic mechanisms involved in addiction is beginning to emerge and the insight gained from these studies will be key to the identification of novel targets for improved diagnosis and treatment of addiction syndromes in humans.

INTRODUCTION

Drug addiction, or substance use disorder (SUD), is defined as the compulsive administration of drugs of abuse in spite of negative consequences to the individual and a loss of control over the intake of such drugs (American Psychiatric Association, 2013). From a neurobiologic perspective, addiction represents maladaptive neural plasticity in response to drugs of abuse, which involves long-term molecular alterations in key brain regions that control reward (Fig. 48.1), leading to potentially lifelong behavioral abnormalities (reviewed in Nestler (2014)). It is estimated that genetic factors contribute ~50% of the susceptibility to addiction for all abused drugs (reviewed in Wang et al. (2012)), with exposure to a range of “risk factors” contributing the other half. These risk factors include adverse early life experiences and other environmental stimuli that “prime” an individual to be more vulnerable to addiction

(Pena et al., 2014). Indeed, without exposure to drugs (an environmental factor itself), the addicted phenotype would not be detected, thus highlighting the complex interplay between genes and environmental factors involved in the initiation and maintenance of the addicted phenotype (Fig. 48.2).

The multifactorial nature of addiction has led to the investigation of both genetic and epigenetic mechanisms underlying the phenotype. Selective breeding studies of strains of rats displaying high and low susceptibility to morphine addiction were the first to confirm that there was a heritable component of addiction (Nichols and Hsiao, 1967). While subsequent studies utilizing selective breeding in animal models have revealed a heritable component for preference to methamphetamine (Wheeler et al., 2009) and ethanol (Gilpin et al., 2008), much of the evidence for a robust genetic contribution

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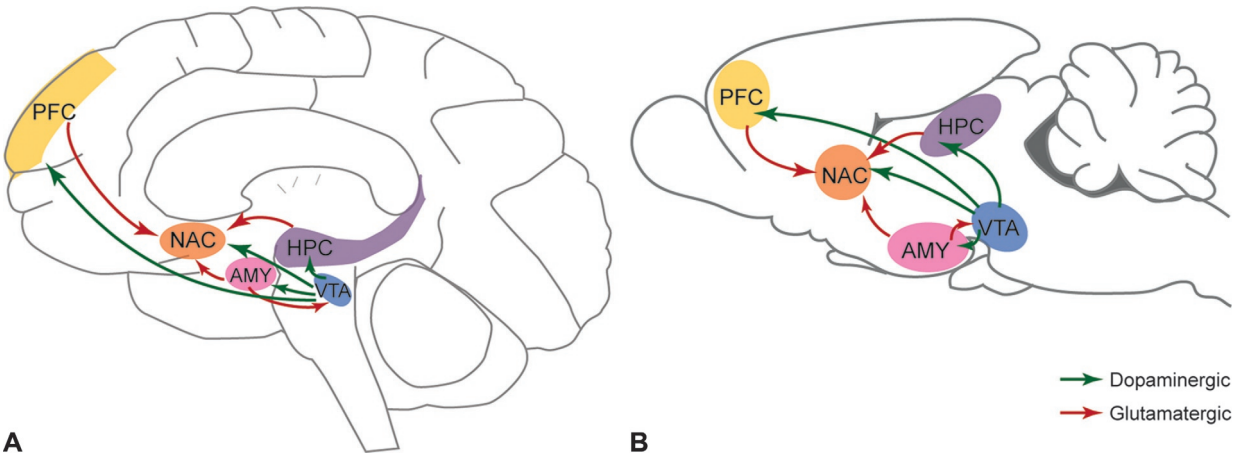


Fig. 48.1. The reward circuitry of the brain is similar across species and is activated by drugs of abuse. The major brain regions involved in the mesolimbic reward pathway are depicted in the human (A) and rodent (B) brain: dopaminergic neurons (green) in the ventral tegmental area (VTA) project to the nucleus accumbens (NAC), prefrontal cortex (PFC), amygdala (AMY), and hippocampus (HPC). The NAC also receives glutamatergic (red) innervation from the PFC, AMY, and HPC. While the mechanisms of action are specific for each drug, most drugs of abuse increase dopaminergic signaling from the VTA to other regions of the reward circuitry. Studies investigating the contribution of genetic factors to the addicted phenotype have focused on identifying markers in vulnerable human subjects that presumably result in altered sensitivity and function of the mesolimbic dopamine system. On the other hand, studies investigating epigenetic mechanisms of drug abuse have focused on the NAC in animal models of addiction, as it is a major region of integration for rewarding stimuli. (Modified with permission from Pena CJ, Bagot RC, Labonte B, et al. (2014) Epigenetic signaling in psychiatric disorders. *J Mol Biol* 426: 3389–3412.)

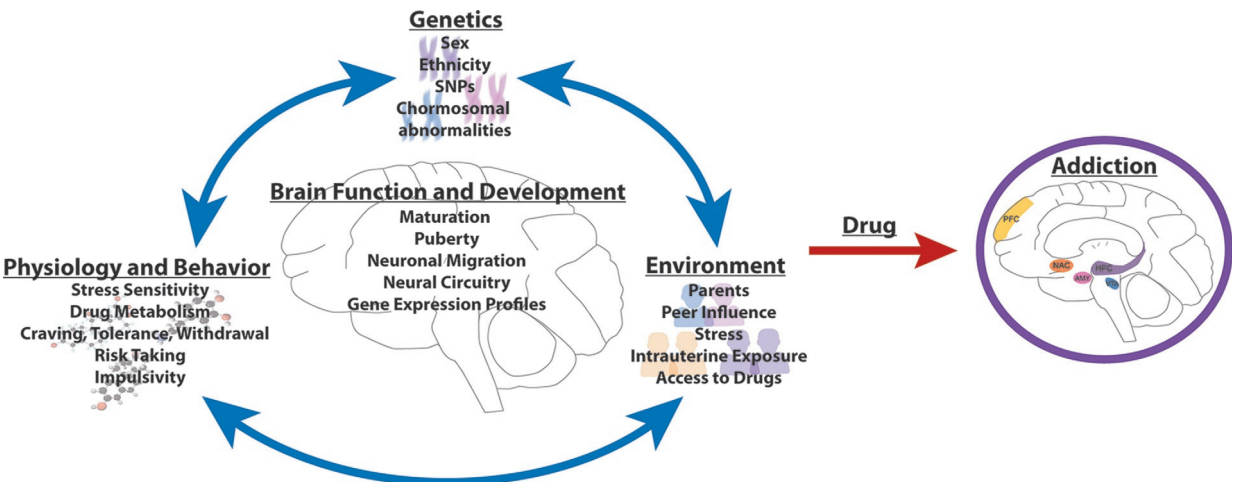


Fig. 48.2. Addiction is a complex phenotype that is regulated by both genetic and environmental factors. Information from the environment is recognized by the brain or body and in turn elicits a response, which often involves changes in gene expression, as indicated by the blue arrows. These gene–environment interactions are relayed by epigenetic mechanisms, including chromatin modifications, DNA methylation, and the expression of noncoding RNAs. Vulnerability to substance abuse has both genetic and environmental risk factors that act in concert to produce the phenotype but exposure to drugs of abuse (indicated with the red arrow) is necessary for the behavioral phenotype to emerge. The details of gene–environment interactions through the full life cycle of addiction are highly iterative and remain incompletely understood. AMY, amygdala; HPC, hippocampus; PFC, prefrontal cortex; SNPs, single-nucleotide polymorphisms; VTA, ventral tegmental area.

to addiction has come from family, twin, and adoption studies in humans. Advances in molecular biology have enabled linkage studies, those comparing genetic markers in related individuals, and association studies, those comparing markers in unrelated individuals, which

initially identified candidate genes and genomic regions weakly involved in addiction. More recent investigations have expanded to genomewide comparisons in humans, which are uncovering more bona fide genetic factors. Therefore, much of the focus on the genetic

component of addiction in this chapter will focus on human studies.

On the other hand, studies focusing on epigenetic mechanisms of addiction have relied primarily on animal models. In its broadest definition, epigenetics describes changes in gene expression that do not arise from changes to the DNA sequence; this generally involves alterations in histone modifications, DNA methylation, and noncoding RNAs (microRNAs (miRNAs) and long noncoding RNAs (lncRNAs); see Chapter 5). Each of these mechanisms can be altered in response to internal and external signals and provides a mechanism by which environmental stimuli can interact with an individual's genome to influence cellular responses and function, including neuroplasticity. The differing temporal effects of each mechanism are, in part, the result of their varying levels of stability – e.g., most histone modifications are exceptionally dynamic, whereas DNA methylation is generally less so (Jaenisch and Bird, 2003). Later in this chapter recent work in the field of epigenetics and addiction will be reviewed, with a major focus on histone modifications, as much of the work examining epigenetic mechanisms involved in the addicted phenotype has focused on such modifications. The much smaller number of studies focusing on the role of DNA methylation and noncoding RNAs are covered briefly as well.

GENETIC COMPONENTS OF DRUG ADDICTION

Family, twin, and adoption studies

Family-based studies examine the risk of substance abuse in first-degree relatives (parents, siblings, offspring) with an SUD and suggest that addiction indeed “runs in the family.” The familial aggregation of alcoholism has been well established (reviewed in McGue (1994)). For example, siblings of ethanol-dependent probands were more likely to be dependent on ethanol when compared to controls (Bierut et al., 1998) and individuals with an alcoholic parent were five times more likely to be alcoholics (Midanik, 1983). While fewer studies have focused on other drugs of abuse, evidence suggests that first-degree relatives of substance abusers have higher rates of addiction when compared to controls (Bierut et al., 1998; Merikangas et al., 1998). Specifically, adult first-degree relatives of probands with an addiction to various drugs, including opiates, cocaine, cannabis, and/or ethanol, displayed an eightfold increased risk of a SUD (Merikangas et al., 1998). Additionally, offspring of substance abusers were twice as likely to develop substance abuse and three times more likely to develop addiction to ethanol or marijuana when compared to children of unaffected parents (Merikangas and Avenevoli, 2000).

One drawback of family-based studies is the difficulty in distinguishing the contribution of genetic variance in individuals vs. environmental factors. Indeed, there is intriguing evidence in animal models for the complexity of this interplay between genes and environment. A recent study revealed that male rats with a history of cocaine self-administration sire offspring that show reduced predilection to self-administer cocaine (Vassoler et al., 2013), while those with a parental history of cannabis exposure show increased predilection to self-administer heroin (Szutorisz et al., 2014). It remains unknown whether such transgenerational transmission of addiction-like vulnerability is mediated behaviorally, through epigenetic modifications in sperm, or through chemical changes in seminal fluid, among other possibilities. In any event, these data raise the possibility that an individual's risk for addiction is mediated not just by genes and environment but also by additional factors that are transmitted across generations.

Adoption and twin studies have been utilized to parse the relative contribution of genetic and environmental factors to the addicted phenotype. Adoption studies compare traits in the adopted offspring with those of the adoptive and biologic parents. While there are difficulties associated with adoption studies (reviewed in Hall et al. (2013)), they have been instrumental in recognizing genetic contributions to the development of addiction by identifying a link between substance abuse in the biologic parent and adopted offspring. Adoption studies have also highlighted environmental factors important for the development of addiction by linking abuse in the adopted offspring to characteristics in the adopted parents. In particular, parental divorce or parental psychiatric disorders were identified as risk factors for the development of substance abuse in offspring (Cadoret et al., 1986, 1995, 1996).

Twin studies, on the other hand, utilize data from monozygotic (MZ) and dizygotic (DZ) twins, who presumably share much of the same environmental influences, to quantify the contribution of heritable factors. If genetics contributes to the development of substance abuse, then outcomes will be more similar between MZ than DZ twins. If environmental factors contribute to the disorder, then outcomes in MZ and DZ will be similar. Evidence from twin studies consistently estimates that the genetic contribution to substance abuse disorders is ~50% for alcoholism (Heath et al., 1993; Prescott and Kendler, 1999; True et al., 1999) and ranges from 30% to 80% for other drugs of abuse (Tsuang et al., 1996; Kendler et al., 2000, 2006; Agrawal et al., 2005; Pergadia et al., 2006). Interestingly, evidence suggests that genetic factors play a greater role in the severity of substance abuse (i.e., the development of addiction) than

in earlier stages of drug use, which are seemingly influenced to a greater extent by environmental factors (Tsuang et al., 1996; Kendler and Prescott, 1998; Rhee et al., 2003; Fowler et al., 2007), highlighting the multifactorial nature of the addiction phenotype.

Linkage and association studies

While twin and adoption studies establish a heritable component, presumably through variations in the DNA sequences of affected individuals, in the acquisition and maintenance of addiction, they do not identify the specific genes/alleles involved. Traditionally, linkage analysis has been used to identify the association between DNA markers and addiction status within individual families. Such studies focus on highly variable regions of the DNA (polymorphic regions) to increase the likelihood of identifying loci – e.g., containing a single-nucleotide polymorphism (SNP) – that are associated with an SUD. Association studies, on the other hand, identify allelic markers in unrelated individuals generally utilizing a case-control experimental design. Unfortunately, few linkage or association studies have identified consistent, replicable markers for SUDs to any of several drugs of abuse (Guerrini et al., 2005; Zhong and Zhang, 2005; Kuo et al., 2006; Clarimon et al., 2007; Gelernter et al., 2007; Hopfer et al., 2007; Li et al., 2007; Loh et al., 2007; Pomerleau et al., 2007; Saccone et al., 2007), thus highlighting the highly polygenic nature of the syndrome. Examples are addiction-associated polymorphisms in the vicinity of dopamine receptors (Drd2, 3, and 4) (reviewed in Gorwood et al. (2012)), GABAergic systems (Noble et al., 1998; Edenberg et al., 2004; Agrawal et al., 2006), and opioid receptors (reviewed in Hall et al. (2013)) in drug dependence, but the effect sizes are very small and only ~50% of all studies report a positive association. It should be noted that many linkage studies have focused on genes that were chosen based on *a priori* hypotheses stemming from neurobiologic mechanisms of addiction, while the genetic risk for addiction likely involves numerous genes and biochemical pathways, as well as nongenic regions of the genome, that have not yet been implicated in addiction. Going forward, it is crucial that candidate genomic regions be identified first via unbiased genomewide studies coupled with follow-up studies in animal models where the genomic loci can be manipulated and their function characterized.

Genomewide association studies

The lack of replication of linkage and association studies of candidate genes has led to the use of genomewide association studies (GWAS) as a superior method to identify common SNPs in SUD individuals. The first

GWAS, which focused on multidrug addiction, was conducted in 2001 (Uhl et al., 2001), but resolution was not sufficient to implicate specific genes until 2005 (Liu et al., 2005). Those original investigations focused on addiction to nicotine (Uhl et al., 2007; Drgon et al., 2009), methamphetamine (Uhl et al., 2008b), ethanol (Uhl et al., 2008a), and multiple drugs (Liu et al., 2006). However, many SNPs initially associated with addiction by GWAS were not replicated in subsequent studies due to the small sample sizes involved (Bierut, 2011). Therefore, recent studies have used aggregate analyses of several thousands of subjects and are beginning to yield, for the first time, robust, bona fide genomic loci that are associated with nicotine, ethanol, or opiate addiction (Saccone et al., 2010; Kapoor et al., 2012; Gelernter et al., 2014a, b), including genes associated with ethanol metabolism (Gelernter et al., 2014a), cholinergic nicotinic receptors (Saccone et al., 2010; Kapoor et al., 2012), and calcium and potassium signaling (Gelernter et al., 2014b). Importantly, very few candidate genes initially hypothesized to be involved in drug addiction have been consistently identified using GWAS, emphasizing the need for more unbiased genomewide studies involving many tens, if not hundreds, of thousands of individuals in the field of addiction.

Conclusions for genetic contributions to drug addiction

It is clear that there is a strong (~50%) heritable component to addiction, which has been supported in family-based, adoption, and twin studies. However, investigations seeking to identify specific genetic markers for addiction in humans, until very recently, have produced inconsistent results. More recent genomewide studies, involving the unbiased analysis of thousands of individuals, are beginning to identify genes that are significantly associated with addiction. However, these identified genes each contribute only a very small amount (typically <1%) to the overall risk, which highlights the tremendous complexity of the phenotype. In this regard, the genetic basis of addiction is similar to that for most other common, complex syndromes (type 2 diabetes, congestive heart failure, psychotic disorders), with perhaps several hundred genes comprising the genetic risk for such a syndrome, like addiction, in any given individual. Identifying those myriad genes and understanding how they combine to increase addiction risk is a major challenge of current research. One approach is to study the interactions of gene networks coupled with transcriptomic analyses in postmortem human brain tissue and animal models (Parikshak et al., 2015). Ongoing efforts are identifying networks that might be instrumental in addiction risk. Therefore,

moving forward, more genomewide studies in combination with advanced bioinformatics techniques are necessary to identify the complex network of genes that determine an individual's vulnerability for addiction.

EPIGENETIC MECHANISMS OF ADDICTION

Because of the strong influence of external risk factors on the development of addiction, it has been proposed that epigenetic mechanisms regulate the long-term changes associated with phenotypic priming and the subsequent addicted phenotype. To date, a majority of the work on the epigenetic mechanisms involved in addiction has focused on the effects of drugs of abuse in animal models of addiction, with a concentration on the effects of stimulants in the nucleus accumbens (NAc), a key brain region involved in reward. Recently, work has expanded to other brain regions in the reward circuitry (Fig. 48.1), including the ventral tegmental area (VTA), prefrontal cortex (PFC), basolateral amygdala (BLA), and hippocampus (HC), as well as to other drugs of abuse. Therefore, the focus of this section will be on stimulant effects in the NAc of animal models, but will include information from other brain regions and drugs of abuse where information is available. It will be important in the future to expand such analyses to human postmortem brain tissue. A related endeavor is to perform epigenetic analyses on human blood or other peripheral tissues of living patients. However, given that drug-induced epigenetic changes in animals show considerable variations among brain regions, it will be crucial to establish that epigenetic signatures in the periphery are predictive of central mechanisms.

Histone modifications

DNA is condensed into the nucleus of the cell in a highly organized and compact manner, referred to as chromatin. The functional unit of chromatin, the nucleosome, is composed of ~147 basepairs wrapped around core histone octamers consisting of two copies of each of the following proteins: H2A, H2B, H3, and H4 (Fig. 48.3 and Table 48.1). Each histone protein can undergo numerous posttranslational modifications in which different functional groups are covalently added to amino acid residues of their N-terminal tails, e.g., acetylation, methylation, phosphorylation, ubiquitination, SUMOylation, citullination, and ADP-ribosylation, among others. These modifications not only alter the structure of the nucleosome but also change the interaction of DNA with the associated histones, thus increasing or decreasing the likelihood of transcription at a given locus. Histone modifications are diverse, and we are only beginning to understand how the various combinations of histone

posttranslational modifications influence or indicate specific transcriptional states (Maze et al., 2014; see Chapter 5). Finally, histone modifications are added or removed by a large family of enzymes, referred to as “writers” and “erasers,” respectively, making them reversible and dynamic epigenetic modifications. While a myriad of histone modifications and their interactions are likely involved in the regulation of the acquisition and maintenance of the addicted phenotype, to date, histone acetylation and methylation are the most extensively studied posttranslational modifications in the addiction field.

HISTONE ACETYLATION

Histone acetylation is generally associated with a permissive transcriptional state (Fig. 48.3). By negating the positive charge associated with the lysine residues on histone tails, acetylation promotes an “open” chromatin state. While the effects of drugs of abuse are widespread, a picture of how histone acetylation is changing throughout the brain after exposure to drugs of abuse, including cocaine (Rogge and Wood, 2013), morphine (Mashayekhi et al., 2012; Wang et al., 2015), nicotine (Levine et al., 2011; Huang et al., 2013, 2014), and others, is beginning to emerge (Table 48.1). In general, acute or repeated exposure to drugs of abuse, including cocaine, ethanol, morphine, and methamphetamine, increases global (i.e., total cellular levels of) acetylation at H3 and/or H4 throughout the reward circuitry, including the whole striatum (Kumar et al., 2005), NAc (Shen et al., 2008; Renthall et al., 2009; Wang et al., 2010; Levine et al., 2011; Botia et al., 2012; Martin et al., 2012; Kennedy et al., 2013), dentate gyrus of the HC (Huang et al., 2014), and BLA (Wang et al., 2015) (Table 48.1). However, these effects may be brain region-specific as ethanol reduces total H3K9ac levels in VTA 14 hours after the last drug exposure (Arora et al., 2013) (for specifics, see Table 48.1).

Global alterations in acetylation provide insight into general changes in transcription in response to drugs of abuse. To understand further the functional outcomes of such changes it is necessary to identify sites of altered histone acetylation at specific genomic regions. Chromatin-immunoprecipitation (ChIP) (Table 48.2) has been used to identify drug-induced increases in pan-acetylation of histone H3 for a number of genes in specific brain reward regions. A common set of genes with enriched histone acetylation, consisting of immediate early genes and those involved in neuroplasticity, is beginning to emerge in the literature (reviewed in Feng and Nestler (2013)) (for specifics, see Table 48.1). While acute and repeated exposures to drugs of abuse have similar effects on global

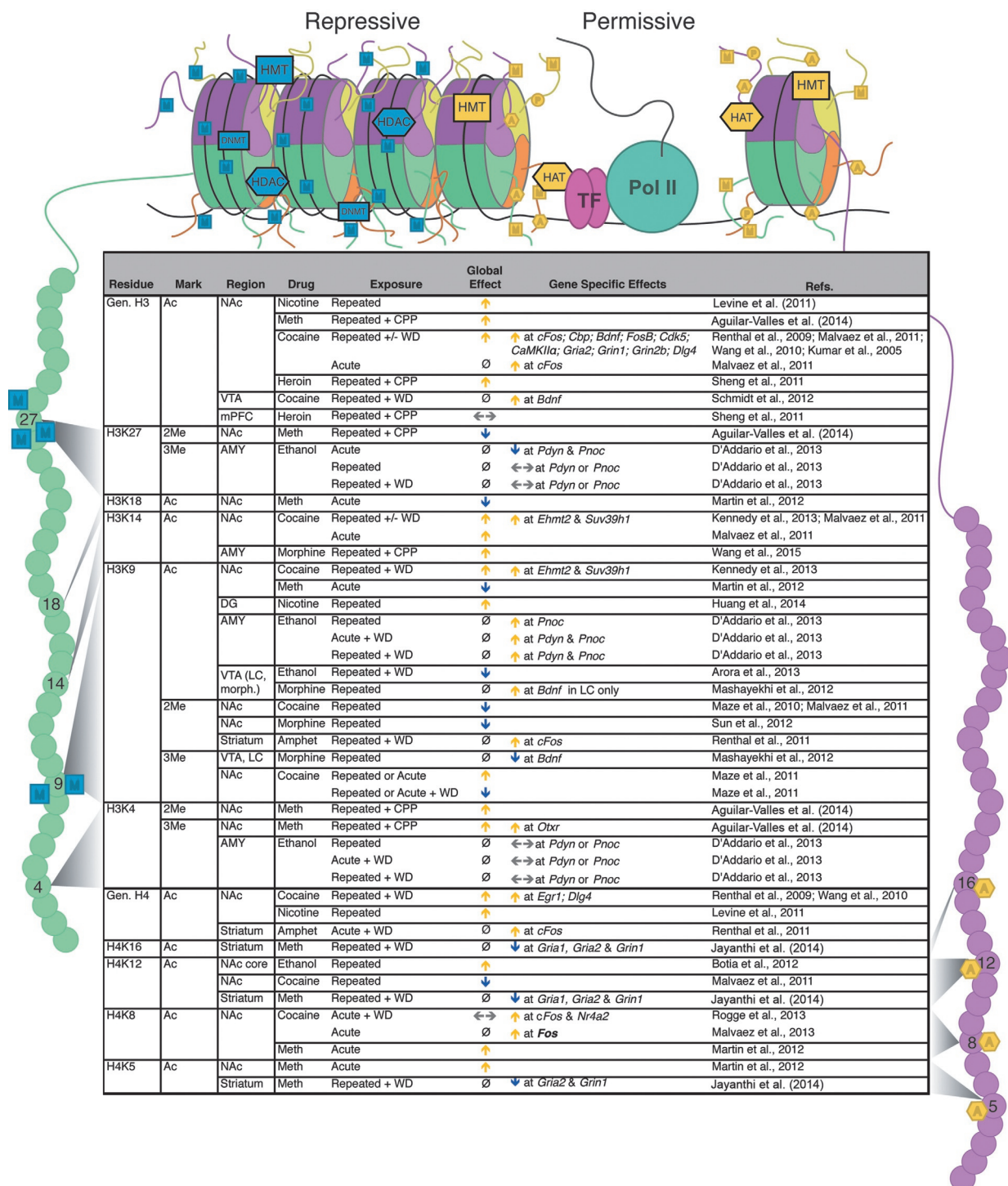


Fig. 48.3 and Table 48.1. Chromatin modifications regulated by drugs of abuse. The illustration (top) indicates histone octamers in a repressive (left) or permissive (right) state. Enzymes involved in maintaining these states and associated transcription factors are indicated and histone tails with specific residues are highlighted as targets of modification: H3 residues subject to methylation or acetylation are indicated in green on the left and H4 residues subject to acetylation are indicated in purple on the right. Table 48.1 lists histone tail modifications of specific residues that are altered in response to drugs of abuse. Arrows indicate an increase (yellow), decrease (blue), or no effect (gray) in specific modifications; ∅ indicates that no information is available. 2Me, dimethylation; 3Me, trimethylation; Ac, acetylation; Amphet, amphetamine; AMY, amygdala; Bdnf, brain-derived neurotrophic factor; CaMKII, calcium/calmodulin-dependent kinase II alpha; Chp, CREB-binding protein; cFos, FBJ murine osteosarcoma viral oncogene

Table 48.2

Methods for studying histone modifications

Method	
ChIP	Involves lightly fixing tissue to cross-link DNA to associated proteins; followed by fragmenting the fixed chromatin by sonication or micrococcal nuclease digestion; and finally, immunoprecipitating the fragmented chromatin with an antibody specific for a given modification of interest (e.g., acetylated or methylated histone)
qChIP	Involves quantifying the amount of immunoprecipitated chromatin at a single genomic region using qPCR analysis
ChIP-chip	Involves quantifying the amount of immunoprecipitated chromatin at thousands of promoter regions using a promoter-specific DNA microarray
ChIP-seq	Involves quantifying the amount of immunoprecipitated chromatin for all genomic regions by high-throughput sequencing ChIP-seq has the additional advantage over ChIP-chip of providing higher basepair resolution for each genomic region
RNA-seq	Involves isolating RNA and converting it to cDNA, followed by high-throughput sequencing RNA-seq has several advantages over older microarray methods, including higher basepair resolution, detection of noncoding RNAs, and the ability to identify alternative splice variants

ChIP, chromatin immunoprecipitation; qChIP, quantitative ChIP; seq, high-throughput sequencing.

Reproduced from Walker et al. (2015).

acetylation levels, histone acetylation at specific loci is dependent on duration of exposure and seemingly reflects functional differences in the role of specific genes in addiction. For example, H4ac is increased at the *cFos* promoter, an immediate early gene, after acute, but not chronic, cocaine treatment, consistent with a desensitization of *cFos* expression only after chronic drug exposure (Kumar et al., 2005; Renthall et al., 2008). In contrast, *Bdnf* and *Cdk5* promoters, genes associated with cell signaling, display enriched H3ac only after chronic drug exposure (Kumar et al., 2005; Mashayekhi et al., 2012; Schmidt et al., 2012), consistent with induction of these genes after chronic exposure to both cocaine (Kumar et al., 2005; Schmidt et al., 2012) and morphine (Mashayekhi et al., 2012). These results suggest that, at the molecular level, the initial response to drugs of abuse is very different from chronic exposures. Understanding the molecular targets of acute and chronic exposures can provide insight into the brain changes associated with acquisition, maintenance, and perhaps extinction of the addicted phenotype and may be helpful in identifying potential

therapeutics and/or biomarkers for different stages or severity of addiction.

Genomewide studies have been instrumental in identifying genes involved in drug-related behaviors and addiction susceptibility. Using ChIP-chip (Table 48.2), an earlier method for analyzing genome-wide changes in chromatin modifications, Renthall et al. (2009) carried out the first genomewide map of pan-acetylation of histone H3 and H4 in the NAc in response to chronic cocaine. The authors reported increases of H3ac or H4ac at 1696 genes in the NAc and reduced levels of acetylation at only 206 genes, consistent with increased acetylation being the predominant modification. Interestingly, there was little overlap of genes displaying alterations of H3ac and H4ac, and most genes did not follow the expected expression pattern predicted by altered histone modifications: hypoacetylation with decreased gene expression and hyperacetylation with increased gene expression (Renthall et al., 2009). These data indicate that additional gene-regulatory mechanisms work in concert with histone acetylation to control drug-induced changes in gene expression.

homolog; *CPP*, conditioned place preference; *DG*, dentate gyrus; *Egr1*, early growth response protein 1; *Ehmt2*, euchromatic histone-lysine *N*-methyltransferase 2; *Gria1*, glutamate receptor, ionotropic, AMPA 1; *Gria2*, glutamate receptor, ionotropic, AMPA 2; *Grin1*, glutamate receptor, ionotropic, *N*-methyl-D-aspartate 1; *Grin2b*, glutamate receptor, ionotropic, *N*-methyl-D-aspartate 2B; *LC*, locus coeruleus; *Meth*, methamphetamine; *mPFC*, medial prefrontal cortex; *NAc*, nucleus accumbens; *Nr4a2*, nuclear receptor subfamily 4, group A, member 2; *Otxr*, oxytocin receptor; *Pdyn*, prodynorphin; *Pnoc*, pronociceptin; *Dlg4*, discs, large homolog 4; postsynaptic density protein 95, PSD-95; *Suv39h1*, suppressor of variegation 3-9 homolog 1; *VTA*, ventral tegmental area; *WD*, withdrawal. (Modified with permission from Vialou et al. (2013).)

Drugs of abuse are thought to alter histone modifications in part by affecting the activity or expression levels of histone acetyltransferases (HATs) and histone deacetylases (HDACs), the enzymes that add or remove acetylation of lysine residues, respectively (Jaenisch and Bird, 2003). Regulation of histone acetylation through manipulation of HAT and HDAC expression has yielded valuable information regarding the role of acetylation in drug-elicited behavioral responses (Feng and Nestler, 2013; Rogge and Wood, 2013) (Table 48.3). In general, decreasing histone acetylation in the NAc either through activation/overexpression of HDACs or inhibition/knockdown of HATs, blunts the behavioral responses to drugs of abuse. On the other hand, increasing histone acetylation in the NAc through pharmacologic inhibition or knockout of HDACs or viral-mediated overexpression of HATs initially facilitates the behavioral effects of many drugs of abuse. However, these effects are sensitive to the timing of HDAC inhibition in relation to exposure to drugs of abuse (Nestler, 2014) (for specifics, see Table 48.3). For example, sustained HDAC inhibition in the NAc can blunt behavioral responses to cocaine by suppressing gene expression through the induction of repressive histone mechanisms (see below) (Kennedy et al., 2013).

Recent studies are beginning to identify more selective effects of individual HDACs and HATs in drug-induced behaviors (for specifics, see Table 48.3). For example, overexpression of HDAC4 (Wang et al., 2010) or HDAC5 (Renthal et al., 2007), or inhibition of CREB-binding protein (a HAT) (Levine et al., 2005; Malvaez et al., 2011), weakens the locomotor response to cocaine (Levine et al., 2005; Malvaez et al., 2011) and acquisition of conditioned place preference (CPP) (Renthal et al., 2007; Wang et al., 2010). Focal deletion of HDAC1, but not HDAC2 or 3, in the NAc mimics the effect of long-term HDAC inhibition by attenuating locomotor responses to cocaine (Kennedy et al., 2013). Interestingly, focal deletion of HDAC3 in the NAc enhances cocaine CPP (Rogge et al., 2013), as does knockout of HDAC5 (Renthal et al., 2007), an effect not observed with HDAC9 (Renthal et al., 2007). The opposing effect observed with HDAC3 could be explained by the finding that HDAC3 plays an important role in drug-induced memory consolidation, as pharmacologic inhibition of HDAC3 enhances extinction of cocaine-induced memories after CPP (Malvaez et al., 2013). Finally, the class III HDACs, termed sirtuins (SIRT), are implicated in the actions of cocaine and morphine (Renthal et al., 2009; Ferguson et al., 2013, 2015). Viral-mediated overexpression of SIRT1 or 2 in the NAc enhances cocaine and morphine reward, whereas the local knockout of SIRT1 has the opposite effect (Ferguson et al., 2013, 2015). This action of SIRT1,

which is opposite to what has been observed with class I and II HDACs, appears to be mediated in large part via the deacetylation of nonhistone proteins, such as the transcription factor, FOXO3a (Ferguson et al., 2015). While it is apparent that differences in histone acetylation state are involved in the behavioral response to acute and prolonged exposure to drugs of abuse, further studies are necessary to identify how these alterations work in concert to produce a temporally specific histone landscape that is indicative of the various stages of addiction.

Taken together, these data support the conclusion that cocaine is a potent regulator of histone acetylation in the NAc (Renthal et al., 2007, 2009). However, histone lysine acetylation is not the only modification contributing to expression changes and the “histone code” regulating these alterations is exceptionally complex (Jenuwein and Allis, 2001), including acetylation of additional residues as well as numerous other types of histone modifications. Further investigations, which focus on additional histone modifications, are necessary to develop a complete understanding of how each modification (or combination of modifications) regulates gene expression in response to drugs of abuse.

HISTONE LYSINE METHYLATION

Histone lysine methylation, another well-characterized histone modification, is associated with either activation or repression of gene expression depending on which residues are methylated and the number of methyl groups added. Indeed, translating the role of histone methylation and its responses to drugs of abuse is much more complex. To date, most of the work on drugs of abuse has focused on the role of repressive histone methylation (Nestler, 2014). However, a recent study in NAc using ChIP-seq coupled with RNA-seq (Table 48.2) investigated how combinations of methylation marks (H3K4me1, H3K4me3, H3K36me3 – all activating marks; and H3K9me2, H3K9me3, H3K27me3 – all repressive marks) correlate with altered transcript levels after exposure to repeated cocaine. While global changes across the genome were not observed, enrichment of H3K4me1 or H3K4me3 at specific gene regions was correlated positively with increased transcription levels, whereas enrichment of H3K9me2, H3K9me3, or H3K27me3 was negatively correlated with transcription (Feng et al., 2014). Other studies have found that repeated exposure to cocaine decreases global levels of H3K9me2 and H3K9me3 in NAc (Maze et al., 2010), whereas repeated opiate exposure (Sun et al., 2012) only decreases H3K9me2 in this brain region. By contrast, methamphetamine decreases H3K27me2 and increases H3K4me2 and H3K4me3 globally in NAc when

Table 48.3

“Writers” and “erasers” of histone modifications mediate behavioral responses to drugs of abuse

Histone Acetylation									
Histone Acetyl Transfrases (HATs)	Expression changes	Time of Exposure	inhibition / knockout	Behavioral Effect of Inhibition	activation / overexpression	Behavioral Effect of Overexpression	Region	Drug	Reference
CBP		Pre-treatment	viral mediated	↓ locomotor sensitization ↓ CPP acquisition			NAc	cocaine	Malvaez, M et al., 2011
Histone Deacetylations (HDACs)	Expression changes		inhibition / knockout	Behavioral Effect	activation / overexpression		Region	Drug	Reference
Class I	HDAC 1	Pre-treatment	viral mediated / pharmacological	↓ locomotor sensitization			NAc	cocaine	Kennedy et al., 2013
	HDAC 2	↓ protein Pre-treatment	viral mediated	↔ locomotor sensitization			NAc NAc	Meth cocaine	Martin et al., 2012 Kennedy et al., 2013
		↑ Protein expression ↑ Protein expression					Nac	Meth	Martin et al., 2012
	HDAC 3	Pre-treatment	viral mediated	↔ locomotor sensitization			VTA	EtOH	Arora DS et al., 2013
		Pre-treatment	viral mediated	↔ locomotor sensitization			NAc	cocaine	Kennedy et al., 2013
		Pre-treatment Concurrent	viral mediated pharmacological (1- 4 doses)	↑ CPP acquisition ↑ CPP extinction ↓ CPP reinstatement			NAc Systemic	cocaine cocaine	Rogge GA 2013 J Neuroscience Malvaez et al., 2013
Class II A	HDAC 8								
	HDAC 4	↔ mRNA Pre-treatment			viral mediated	↓ drug motivation ↓ CPP acquisition	NAc	Cocaine	Wang L et al., 2010 Neuropsychopharmacology; Kumar et al., 2005; Rogge et al., 2013
	HDAC 5	↓ mRNA ↔ mRNA Pre-treatment	Knockout organism wide	↑ CPP	viral mediated	↓ CPP acquisition	NAc	Cocaine	Renthal et al., 2007; Rogge et al., 2013
Class IIB	HDAC 7								
	HDAC 9		Pre-treatment		viral mediated	↔ CPP acquisition	NAc	Cocaine	Renthal et al., 2007
	HDAC 6								
	HDAC 10	↔ mRNA					Striatum	EtOH	Botial et al., 2012

Continued

Table 48.3

Continued

Histone Deacetylases (HDACs)		Expression changes		inhibition / knockout	Behavioral Effect	activation / overexpression		Region	Drug	Reference
Class III	Sirt 1	↑ mRNA & protein	Pre-treatment	viral mediated	↓ CPP Acquisition	viral mediated	↑ CPP acquisition & Locomotor Sensitization	NAc	Cocaine	Renthal et al., 2009; Ferguson et al., 2013
		↑ mRNA & protein						NAc	Morphine	Ferguson et al., 2013
	Sirt 2	↑ mRNA & protein	Pre-treatment			viral mediated	↑ CPP acquisition	NAc		Renthal et al., 2009 Ferguson et al., 2013
	Sirt 3									
	Sirt 4									
	Sirt 5									
	Sirt 6									
Class IV	Sirt 7									
	HDAC 11	↓ mRNA						Striatum	EtOH	Botial et al., 2012
General HDAC Inhibitors		Expression changes		# of doses HDACi	Behavioral Effect	# of doses agonist		Region	Drug	Reference
	TSA		Pre-treatment	3	↓ hyperactivity			NAc BLA	morphine heroin morphine	Jing L et al., 2011 Sheng J, et al., 2011; Wang Y et al., 2015; J Mol Neuroscience
			Concurrent	5 ~30	↑ CPP acquisition ↑ CPP acquisition ↑ CPP extinction ↓ CPP					
	VPA		Pre-treatment	7	↓ reinstatement ↓ locomotor sensitization			micro injection into NAc	amphetamine	Kim WY et al., 2008
			Concurrent	8	↑ locomotor sensitization			systemic	amphetamine	Shen H-Y et al., 2008; Kalda et al., 2008
			Post-treatment	6	↓ locomotor sensitization			systemic	amphetamine	Kalda et al., 2008
	NaB		Concurrent	1	↑ CPP acquisition ↓ extinction			Systemic	cocaine	Itzhak Y et al., 2013;
				2	↑ locomotor sensitization			Systemic	cocaine	Kumar et al., 2005
				10	↓ locomotor sensitization			Systemic	cocaine	Schroeder FA et al., 2008
			Post-treatment	1	↑ self-administration			systemic	cocaine	Sun J et al., 2008
				4	↓ CPP extinction ↔ CPP			systemic	cocaine	Itzhak Y et al., 2013;
				7	↑ reinstatement ↑ CPP extinction ↓ CPP reinstatement			systemic		Malvaez M et al., 2010

Table 48.3

Continued

General HDAC Inhibitors	Expression changes		# of doses HDACi	Behavioral Effect	# of doses agonist	Region	Drug	Reference
TSA		Concurrent	2 7	↑ CPP acquisition ↓ break-point ↓ self-administration		Systemic systemic	cocaine cocaine	Kumar et al., 2005 Romieu P et al., 2008
		Post-Treatment	5	↑ dose response curve ↑ break point		NAc shell microinjections	cocaine	Wang L et al., 2010
Sirtinol		Pre-treatment	~10	↓ CPP acquisition ↓ self-administration		NAc osmotic mini pump	cocaine	Renthal et al., 2009
Resveratrol		Concurrent	4	↑ CPP acquisition		systemic	cocaine	Renthal et al., 2009
Histone Methylation								
Histone Methyltransferases (HMTs)	Expression changes		inhibition / knockout	Behavioral Effect	activation / overexpression	Region	Drug	Reference
G9a	↓ mRNA expression	Pre-treatment	viral mediated / pharmacological	↑ CPP acquisition		NAc	cocaine/ morphine	Maze I et al., 2010; Sun H et al., 2012
GLP	↓ mRNA expression	Pre-treatment				NAc	cocaine/ morphine	Maze I et al., 2010; Sun H et al., 2012
MLL1 (Mixed-lineage leukemia 1)	↑ mRNA in CPP ↓ mRNA in home cage	Pre-treatment	viral mediated	↓ CPP acquisition		NAc	Meth	Aguilar-Valles, A et al., 2013
Histone Demethylases (HDMs)								
Histone Demethylases (HDMs)	Expression changes		inhibition / knockout	Behavioral Effect	activation / overexpression	Region	Drug	Reference
histone lysine (K)-specific demethylase 5C (KDM5c)	↔ mRNA expression	Pre-treatment	viral mediated	↓ CPP consolidation		NAc	Meth	Aguilar-Valles, A et al., 2013

The expression changes, brain region, drug of abuse, timing of exposure, type of manipulation, and behavioral outcome are listed for each enzyme. Only those enzymes with available information are included. Pre-treatment indicates that the manipulation occurred at least 24 hours before exposure to drug of abuse. Concurrent indicates that the manipulation occurred between 60 and 0 min before exposure to drug. Post-treatment indicates that the manipulation occurred after exposure to drug of abuse. Arrows indicate an increase (yellow), decrease (blue), or no effect (gray).

Amphet, amphetamine; BLA, basolateral amygdala; CBP, CREB-binding protein; CPP, conditioned place preference; G9a, euchromatic histone-lysine N-methyltransferase 2; GLP, G9a-like protein; HC, home cage; KDM5c, lysine (K)-specific demethylase 5C; Meth, methamphetamine; MLL1, myeloid/lymphoid mixed lineage leukemia 1; NAb, sodium butyrate; NAc, nucleus accumbens; TSA, trichostatinA; VPA, valproic acid; VTA, ventral tegmental area.

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exposure is coupled with CPP but not in the animals' home cage (Aguilar-Valles et al., 2014), again suggesting that regulation of histone methylation in response to drugs of abuse is complex and likely drug- and region-specific (Table 48.1).

As mentioned above, an emerging theme developing in the field suggests that inhibition of repressive epigenetic modifying enzymes (e.g., class I and II HDACs) and those histone methyltransferases (HMTs) that mediate gene repression, in the NAc enhances drug-associated behaviors. Studies manipulating these HMTs and associated histone demethylases (HDMs) support this. A recent study investigating mixed-lineage leukemia 1 (MLL1) and histone lysine (K)-specific demethylase 5C (KDM5c), which catalyze the methylation and demethylation of H3K4me3, a permissive mark, respectively, implicate H3K4me3 in drug-elicited behaviors. Methamphetamine administration increases expression of *Mill* mRNA in NAc when paired with a novel context; however, when exposure occurs in the home cage, expression is decreased. NAc-specific knockdown of MLL1 reduces global H3K4me3 levels and inhibits methamphetamine-induced CPP. Interestingly, NAc-specific knockdown of KDM5c increases global H3K4me3 levels and inhibits methamphetamine CPP if the knockdown occurs after consolidation (between test days 1 and 2) (Aguilar-Valles et al., 2014). This suggests that the methylation state of H3K4 in NAc plays a dynamic role in the temporal- and context-specific neuroplasticity required for development and expression of methamphetamine-induced behavioral adaptations.

Methylation of H3K9 is also critical in regulating drug-induced behavioral responses. Two HMTs for H3K9me2, G9a and G9a-like protein (GLP), are down-regulated in the NAc by chronic cocaine (Maze et al., 2010, 2011) or opiates (Sun et al., 2012), which is associated with global decreases in levels of H3K9me2. This effect was not observed for other H3K9/K27 HMTs (Maze et al., 2010; Sun et al., 2012). Pharmacologic inhibition or viral-mediated knockdown of G9a facilitates cocaine- or opiate-mediated CPP, whereas overexpression opposes drug-induced effects (Maze et al., 2010; Sun et al., 2012) (Table 48.3). Indeed, the paradoxical suppression of cocaine-elicited behaviors after prolonged HDAC inhibition in NAc, or local knockout of HDAC1 (see above) appears to be mediated via the induction of G9a and its consequent repression of gene expression (Kennedy et al., 2013) (Table 48.1). Taken together, these data suggest that histone methylation plays an important and complex role in drug-induced behaviors.

As noted above, chronic cocaine administration causes decreases in global levels of H3K9me3 in NAc (Maze et al., 2011). This mark, along with H3K9me2, is thought to contribute to the silencing of genes in

heterochromatic regions. As might be expected from the overall depletion of these marks and a reduction in the amount of heterochromatin, we observed an increase in the size of neuronal nuclei in NAc, an effect not seen in other brain regions, after chronic cocaine administration. ChIP-seq (Table 48.2) for H3K9me3 found most of the binding in nongenic regions, where its depletion after cocaine treatment was associated with increased expression of certain repetitive elements (Maze et al., 2011). These data highlight the highly complex genomic changes that a drug of abuse induces in target neurons in the brain and defines further studies of nongenic regions in future years.

OTHER HISTONE MARKS

Few studies have investigated the role of other histone modifications in response to drugs of abuse; however, a recent study suggests that polyADP-ribosylation of histones, generally a permissive mark, plays a role in cocaine-induced neuronal plasticity. Expression of polyADP-ribosyltransferase-1 (PARP1), a "writer" of histone polyADP-ribosylation, is increased in the NAc after repeated cocaine exposure, leading to an increase in histone polyADP-ribosylation. Furthermore, overexpression of PARP1 in NAc enhances cocaine locomotor sensitivity and cocaine-induced CPP, as well as cocaine self-administration, while infusion of a PARP1 inhibitor or overexpression of polyADP-ribosylglycohydrolase (PARG), an "eraser" of histone polyADP-ribosylation, yields opposing results. After repeated cocaine, there is a genomewide increase of PARP1 binding across genes, as determined by ChIP-seq (Table 48.2), and a significant correlation of greater PARP1 enrichment with increased gene expression (Scobie et al., 2014). These results are consistent with a permissive transcriptional environment following repeated cocaine exposure identified in previous studies. Further investigation of other histone modifications is under way, but in the future it will be necessary to integrate genomewide alterations in a multitude of histone modifications in order to identify the complex code that is indicative of altered gene expression and an addicted phenotype.

DNA methylation

An important epigenetic regulator in addition to histone modifications is DNA methylation, which occurs with the addition of a methyl group to cytosine-phosphoguanine (CpG) at the C5 position (5-mC). In development, it plays a pivotal role in tissue-specific gene expression, cellular differentiation, X inactivation, imprinting of parental alleles, and repetitive element silencing. DNA methylation at gene promoters is generally associated with repression, while methylation in gene bodies has been associated with active transcription in several species, including humans (Suzuki and Bird, 2008). Another cytosine

modification, 5-hydroxy-mC (5-hmC) appears to be concentrated in gene bodies (Tahiliani et al., 2009), is enriched in brain (Kriaucionis and Heintz, 2009), and associated with gene regulation (Szulwach et al., 2011). While DNA methylation has traditionally been characterized as a stable epigenetic mark, recent evidence suggests that it is much more transient than previously thought (Mikaelsson and Miller, 2011). It is clear that DNA methylation and histone modifications participate in an “epigenetic conversation” to regulate transcription. For example, DNA methylation can recruit methylation-binding proteins, which in turn recruit HDACs, leading to a repressed chromatin state (Hashimoto et al., 2010).

There is a dearth of information regarding the role of DNA methylation in drug addiction but, of the few studies that exist, many are focused on how drugs of abuse regulate expression of DNA methyltransferases (DNMTs), the enzymes that catalyze the addition of methyl groups to DNA. Those studies suggest that there are dynamic expression changes in response to drugs of abuse. In general, both acute exposure to cocaine and prolonged withdrawal (28 days) from chronic exposure increase the expression of *Dnmt3a*, the enzyme regulating de novo DNA methylation in mammals, in the NAc (LaPlant et al., 2010; Anier et al., 2013). Studies investigating repeated exposure to cocaine have produced conflicting results (Anier et al., 2010, 2013; LaPlant et al., 2010). In the PFC, *Dnmt3b* expression is downregulated 2 hours after cocaine but not morphine CPP, an effect not observed in the NAc (Tian et al., 2012). Additionally, cocaine but not morphine decreases global DNA methylation levels in the PFC, which can be rescued with the methyl donor methionine (Tian et al., 2012). Interestingly, treatment with methionine inhibits cocaine and food CPP, but had no effect on morphine CPP (LaPlant et al., 2010; Tian et al., 2012), thus demonstrating both the drug- and regional specificity of this epigenetic mechanism.

Studies have identified a role for DNA methylation in drug-elicited behaviors by manipulating targets associated with DNA methylation in the brain. Pharmaceutical inhibition or local knockout of DNMT3a in the NAc enhances cocaine reward, whereas overexpression decreases it (LaPlant et al., 2010). Supplementation with methionine, a methyl donor which increases DNA methylation overall, also attenuates behavioral responses to cocaine, including its self-administration (Wright et al., 2015). Additionally, NAc-specific knockout of methyl CpG-binding protein 2 (MeCP2), a methyl-binding protein and important modulator of neural plasticity, enhances amphetamine reward (Deng et al., 2010), whereas cocaine increases expression of MeCP2 specifically in the dorsal striatum and local knockdown of the protein inhibits cocaine self-administration

(Im et al., 2010), suggesting again that repression of transcription either through histone modification or DNA methylation generally blunts the actions of drugs of abuse.

Recent work has demonstrated that enzymes involved in the generation of 5-hmC also influence cocaine action (Feng et al., 2015). Chronic cocaine administration downregulates the expression of ten eleven translocation-1 (TET1) in NAc. TET1 oxidizes 5-mC into 5-hmC and ultimately, with the involvement of additional enzymes, into nonmethylated cytosine. This suppression of TET1 expression serves to enhance behavioral responses to cocaine.

While it is intriguing that the enzymes and other proteins involved in regulating DNA methylation are altered by drugs of abuse, there has still not been a global mapping at single-nucleotide resolution of DNA methylation changes in the reward circuitry in response to drugs of abuse, although a few studies have found gene-specific effects on 5-mC or 5-hmC in the brain (Taqi et al., 2011; Nielsen et al., 2012; Ponomarev et al., 2012; Day et al., 2013; Feng et al., 2015; Massart et al., 2015). Unfortunately, due to technical limitations, it remains very expensive to identify altered DNA methylation on a genomewide scale at single-basepair resolution. Additionally, most genomewide assays focus on the promoter regions of genes and provide little information regarding methylation changes in gene bodies or other genomic regions. Given that methylation is predicted to have different functions in regulating gene expression depending on its location, and different cytosine modifications (5-mC and 5-hmC) are often associated with opposing functional outcomes, it is imperative that genomewide maps of DNA methylation with single-base resolution are generated to develop a comprehensive understanding of the role of this epigenetic mark in addiction.

MicroRNAs and long noncoding RNAs

Recent evidence from the complete sequencing of multiple mammalian genomes has revealed a large number of RNA transcripts that are not translated into protein. Many of these noncoding RNAs are now known to play an important role in cell function. Specifically, miRNAs are short sequences of RNA (~20 bp) that exert a repressive role on gene expression by binding a target sequence on specific mRNAs and blocking translation or inducing degradation. LncRNAs are longer sequences of RNAs (>200 bp) that have been found to regulate gene expression by forming RNA-protein complexes and modulating chromatin-regulatory proteins or transcription factors (O'Carroll and Schaefer, 2013).

The role of miRNAs is understudied in the field of addiction (Bali and Kenny, 2013). However, cocaine has been shown to increase expression of miR-181a and decrease miR-124 and let-7d in the striatum of

rodents (Chandrasekar and Dreyer, 2009; Schaefer et al., 2010). These miRNAs in turn influence cocaine-induced behaviors (Chandrasekar and Dreyer, 2011). Morphine increases expression of miR-339-3p in mouse HC, which then targets and decreases expression of the μ -opioid receptor (Wu et al., 2013). It also increases expression of let-7 in the mouse brain (He et al., 2010) and decreases expression of miR-133b in cultured rat hippocampal neurons (Sanchez-Simon et al., 2010), suggesting that there are drug-specific effects with regard to miRNA regulation as well. Manipulation of specific miRNAs in the brain can affect drug-elicited behaviors. For example, cocaine induces the expression of miR-212 in the dorsal striatum after self-administration, and overexpression of miR-212 inhibits cocaine intake (Chandrasekar and Dreyer, 2011). A recent study found that miR-206 is upregulated in the PFC but not the VTA, amygdala, or NAc of rats self-administering ethanol, and overexpression of miR-206 in the PFC in nondependent animals escalated ethanol consumption. Interestingly, a concomitant decrease in expression of *Bdnf*, a predicted target of miR-206, was observed in these animals (Tapocik et al., 2014). Finally, argonaute-2, which is crucial in the processing of miRNAs, has been shown to control behavioral responses to cocaine via actions in *Drd2*-containing striatal neurons (Schaefer et al., 2010). Taken together, these data provide intriguing evidence that miRNAs play an important role in the acquisition and maintenance of addiction.

lncRNAs have not yet been characterized in addiction models. However, there is promising evidence from mining microarray data indicating that lncRNAs are altered in the brains of heroin-addicted humans (Michelhaugh et al., 2011). Now that lncRNAs are being captured far more completely in RNA-seq (Table 48.2) experiments, it is expected that an improved understanding of their contribution to addiction-related neural and behavioral abnormalities will be forthcoming in the near future.

Conclusions for epigenetic contributions to drug addiction

It is clear that epigenetic mechanisms, including histone modifications, DNA methylation, and noncoding RNAs, all play an important and complex role in the neuroplastic changes associated with addiction. Currently, histone acetylation and methylation are by far the most comprehensively studied histone modifications in the field of addiction. A comprehensive review of the data, which includes numerous drugs of abuse and brain regions, reveals several emergent properties with regard to histone modifications (Table 48.1) suggests that: (1) drugs of abuse generally increase the likelihood of transcription in NAc by increasing histone modifications associated

with an “active” chromatin state and reducing those associated with a repressive chromatin state; (2) these activating marks are associated with immediate early genes and many others that have been implicated generally in neuroplasticity; (3) drugs of abuse likely affect the expression or activation of the “writers” and “erasers” of histone modifications, and manipulation of these enzymes in brain regions associated with reward alters the behavioral response to drugs of abuse; and (4) considerable drug-induced epigenetic regulation occurs in nongenic regions of the genome, which is also likely important in mediating the addiction phenotype. While the general principle of a permissive epigenetic-transcriptional state holds for the majority of regulated genes, it is important to emphasize that hundreds of genes show regulation in the opposite direction.

Although data are promising that DNA methylation and noncoding RNAs also play an important role in the establishment of addiction, these endpoints remain understudied such that it is difficult to identify consistent effects of drugs of abuse on these parameters at this time. However, DNA methyltransferases are targets of cocaine and morphine and alterations of these enzymes affect drug-induced behaviors. A better understanding of the temporal and regional alterations of these enzymes, as well as genomewide maps of 5-mC and 5-hmC in brain reward regions, is now necessary. Likewise, we are beginning to identify several specific miRNAs and their targets, which are altered by exposure to drugs of abuse.

The potent influence of epigenetic mechanisms on the addiction process raises the question of whether drugs that target such mechanisms might be useful in the treatment of addictive disorders. The major concern with this approach is that virtually all epigenetic enzymes and related proteins expressed in the brain are also expressed in peripheral tissues, such that drugs that target such mechanisms might be associated with unacceptable side-effects. However, this is ultimately an empiric question and worthy of direct investigation. Several HDAC and HDM inhibitors are being developed by the pharmaceutical industry for the treatment of cancer. It would be interesting to test tool compounds, known to penetrate the brain, in proof-of-concept studies of addiction. Beyond targeting epigenetic mechanisms directly, there is every expectation that the genomewide characterization of a host of epigenetic modifications in discrete brain regions will point the field toward particular genes and biochemical pathways that should be targeted in future drug discovery efforts.

CONCLUDING REMARKS

It is clear that genetic and environmental factors contribute to drug addiction, and adoption and twin studies have

been instrumental in quantifying the contribution of both factors to the syndrome in humans. While there is substantial evidence that genetic factors contribute roughly half of the risk for an addicted phenotype, specific genes and other genomic regions that mediate this risk have been very difficult to identify with certainty. More powerful experimental methods and associated bioinformatic approaches that are agnostic in nature – e.g., GWAS with whole-exome or whole-genome sequencing – are now enabling far more penetrating studies which should reveal gene networks that underlie addiction risk. On the other hand, environmental risk factors are also important and a few vague factors (e.g., child abuse, adverse life experiences) are known, but their mechanisms remain poorly understood, particularly in humans. Epigenetic mechanisms have been studied mainly in animal models of addiction and are beginning to offer an understanding of how environmental stimuli can alter gene expression in the brain on a long-term basis. In the future, it will be necessary to integrate these studies with individual variations in genome sequences in an effort to understand how an individual's ultimate risk of (or resistance to) an addiction syndrome is mediated. The integration of such data should provide us with the ability to identify biomarkers for susceptibility and resilience to addiction and, we hope, lead to the identification of novel therapeutic targets for improved treatment of addiction syndromes.

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